



BUILD BLOCK BRIEF · FALL 2026-27

Machine & Language STUDIO LANE

Planning brief session 4 · build from session 8 · 20 students · 4 teams · courses: GCAA444, GCCA445, GCCE691, GCOS443, GCTA441, GCCN555, GCCN558, GCOS446

What your team is building

Systems-level computing with a language or network dimension — NLP, OS, architecture, networks, automata, algorithms — built and evaluated on a public problem.

ONE PROJECT, SEVERAL GRADEBOOKS

You submit **one** piece of work, and each course that marks it looks at a different dimension, out of 20. **At most eight courses mark one team file.** A course you take that falls outside those eight sets you a **lens note** instead — an individual piece on the same project from that course's angle, 3,000 words, same rubric, same 20 marks. No student carries more than two: from a third, that course marks your viva out of 30 instead.

YOUR MARKS FROM THIS PROJECT

30 the project — a team mark of 20 from each course that marks your file, plus your own viva of 10, held in class in session 14
30 the mid-term
40 the final
The three quizzes are practice and carry no marks.

This replaces coursework, not classes.

What each course marks

CODE	COURSE	WHAT THIS COURSE MARKS — 20 MARKS
GCAA444	Analysis of Algorithm Mariha Fahad	This course marks the cost account: the asymptotic analysis of the algorithms in the study, the design of the benchmark, the measured results, and the honest reconciliation of predicted with observed running time on the team's chosen workload. 20 marks, team.
GCCA445	Computer Architecture Maaz Ahmed	This course marks the hardware account: the benchmark run on the team's program, the cache and pipeline behaviour measured, the architectural change proposed and simulated, and whether the speed-up claimed survives Amdahl's law. 20 marks, team.
GCCE691	Natural Language Processing Haris Yaseen	This course marks the language component of the study: the corpus and how it was prepared, the model built, the evaluation design, the error analysis, and the honesty of the claims made about what the component can and cannot do. 20 marks, team.



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CODE	COURSE	WHAT THIS COURSE MARKS — 20 MARKS
GCOS443	Operating Systems Yahya Ali Rashidi	This course marks the operating-system account: the processes and threads the team's program creates, its synchronisation and whether it is free of races, the scheduling and paging behaviour measured under load, and the system-call trace read and explained. 20 marks, team.
GCTA441	Theory of Automata Ume Farwa	This course marks the formal model: the automaton or grammar written for the team's input language, its conversion and minimisation, the proof that the class chosen is the right one, and the recogniser built from the model. 20 marks, team.
GCCN555	Computer Networks Adeel Karim	This course marks the network: the requirements read out of the brief, the address plan, the device configuration and how it was verified, the packet captures that prove the design works, and the fault log showing what broke and how it was found. 20 marks, team.

What to hand in

- **GCAA444 Analysis of Algorithm** — A measured systems study submitted by session 13 — the workload, the code, the benchmark harness, the predicted and observed costs set out in one table, and the written reconciliation — filed with AI declaration.
- **GCCA445 Computer Architecture** — By session 12, a simulator study of the team's program — baseline CPI and cache statistics, one architectural change implemented in the simulator, before-and-after measurements and a cost argument — with AI declaration.
- **GCCE691 Natural Language Processing** — A measured study submitted by session 13 — corpus, code, evaluation harness, a results table with baselines, and a written error analysis naming the failure cases — filed with AI declaration.
- **GCOS443 Operating Systems** — By session 12, an instrumented build of the team's program with a systems report — trace and perf output, a scheduling and page-fault profile under three workloads, and the synchronisation argument — with AI declaration.
- **GCTA441 Theory of Automata** — By session 13, the formal-language specification for the team's system — grammar or automaton with a state diagram, the minimisation, a proof of class membership, and a working recogniser — with AI declaration.
- **GCCN555 Computer Networks** — A working network by session 13 — simulator project file, address plan, device configurations, annotated packet captures for three key flows, and a test and fault report — submitted together with AI declaration.



Greenwich University



A SCHOOL OF GREENWICH UNIVERSITY

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How big the file is

The written file is **3,000 words for every member**, plus or minus ten per cent — 15,000 words for a team of five, 12,000 for four, 9,000 for three — plus the artefacts: the evidence or data log with its instrument, interview notes and consent log, any program with its tests, the pitch deck, the AI declaration and the draft-to-final revision log. In the venture and client lanes the live thing itself is the largest artefact. Each member owns one section of 3,000 words with their name on it, one artefact they built, and one part they can defend alone at viva. A lens note is 3,000 words on the same project. The full format is in "The Build Block File — required format" in the kit.



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Dates

Session 4	28 Sep	Planning brief issued — twenty minutes of class: the brief, the Ideas Book and the format sheet. Teams locked since session 3. Choose the project, design the instrument or find the client, split the sections.
Guest session	Session 4, 5 or 6	A practitioner from outside the university speaks to your school, on whichever of those three sessions the speaker can make. Attendance is compulsory. Each student writes one page on it for their own file; it is not marked.
Session 7	19 Oct	One-page plan uploaded to the portal: the question, the instrument or the client charter, who owns what. Not marked — it is the gate. This week the convener merges the marking courses' requirements into the file's sections, so the team writes one document and not eight. Clients matched to teams.
Session 8	2 Nov	Build starts. Rubric issued, roles confirmed, team charter uploaded. Build-with-AI workshop for venture and client teams.
Session 9	9 Nov	Scope agreed with the convener. Client or subject confirmed. Your school arranges its industrial visit for some point between now and session 12.
Session 11	23 Nov	Draft review, inside a normal class: each course teacher sees the teams in their own section for about ten minutes each — show what exists, not what you plan. The convener samples three files per block.
Session 12	30 Nov	The artefact goes live or reaches its first complete state.
Session 13	7 Dec	Final file uploaded with its AI declaration. Venture teams evidence the first paying customer. Five of the twenty come off for each week the file is late, in every course marking it, and nothing is accepted after the Karachi Review.
Session 14	14-19 Dec	Vivas in class: each course holds its own, team by team, in its session-14 class — about three minutes per student, on your own part as that course sees it.
The Karachi Review	Tue 15 Dec	Your team exhibits at its own table, across the campus, all day. The day opens with a thirty-minute handover of studies to the public bodies who asked for them; press and industry attend. Ten panels — a teacher and an alumnus each — circulate the exhibition and score you at your table, about eight minutes a team. Not marked separately; the panel sheet goes to every course marking the file.
Weekend	—	Weekend cohorts hold all of the above in their Saturday and Sunday slots; the Karachi Review runs on a weekend afternoon.

Rules for this lane



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What counts as done

A finished piece of work — not a plan for one, not a prototype that only runs on your machine — with the process behind it documented.

Craft

The standard is what a competent professional would accept, at the level of your year. Unfinished work presented as finished is marked as unfinished.

Sources and rights

Anything you did not make yourself is credited. You may not use material you have no right to use, and asset licences are recorded in the file.

If your work involves people

Consenting adults only, with an information sheet, a consent form and a debrief. Mock cases where the subject is clinical. No real patient, client or examinee material, ever, and none of it into an AI tool.

Marks

Craft carries the weight, but the reasoning behind your choices carries the rest. Be able to say why you did it that way.

Using AI

You are expected to use Claude and similar tools — to research, to build, to draft, to model numbers. Three rules. Check everything a tool gives you before you use it, and record the check. Declare what you used in the AI declaration bound into the file, as the HEC *Framework on the Use of Generative AI Tools* (2026) requires. Be able to explain, at the viva, everything your team built. **Work you cannot explain is not credited to you.** Undeclared use, or use to fabricate data, references or results, is academic misconduct. Never put real client, patient or examinee data, or anything identifying a real person, into a tool.

BLOCK CONVENER

PROGRAMME HEAD

DATE ISSUED